

Nebico Oil Retention Architecture

Commissioning Verification Checklist, Rev. 3

Program 3.3, Coolmay L10S · HMI: TK8100H (supervisory) + TK8050H ×3 (per-station) · Prepared by
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How to Use This Document

This checklist is used once Program 3.3 has been written and loaded to the PLC and the TK8100H/TK8050H HMI screens are live. I work through it with the commissioning technician, section by section, on the live system (or a faithful bench/simulation rig), and we tick each item before signing off.

Tolerance Policy

- **Calculated / analog quantities** (temperatures, tank scores, durations, mass proxies, derived setpoints such as T_{target} and T_f) are allowed a $\pm 5\%$ deviation between the hand-calculated expected value and the PLC's live value before a test is marked Fail.
- **Digital / logical behaviour** – valve open/closed state, coil SET/RST, latch behaviour, sequencing, mutual exclusivity, which branch of an IF/ELSE fires, HMI password gating – must match EXACTLY. There is no tolerance on logic; a state is either correct or it is not.
- Items marked **Unsatisfactory** are not live pass/fail tests. They are open items – either a feature described to me that is not present in the Program 3.3 ladder as I have documented it, or a point the PLC Guide does not give an exact register/output for – that must be discussed with the design authority and the resolution written against the item (or an attached deviation note) before the section is signed off.

Marking Results

- **Pass** – behaviour/value matches the expected result within the stated tolerance.
- **Fail** – behaviour/value does not match. Record the observed value/behaviour in an attached deviation note and raise it with me before re-test.
- **Unsatisfactory** – the test point does not apply as stated, could not be verified as specified, or the exact register/output it depends on is not confirmed in the current PLC Guide (e.g. a feature not yet implemented in the ladder, or an item pending design-authority confirmation).

Reference Codes

“PLC Step n ” refers to the corresponding line-statement step, per the Program Structure Overview, of the *Nebico Oil Retention – PLC Guide, Program 3.3*. Where a PLC Step is not given against an item, the PLC Guide does not name an exact coil or register for that point, and the item is treated as open – see the Tolerance Policy above and the Open Items list at the end of this document.

Revision 3. Every PLC Step reference below has been renumbered against the Program 3.3 export (the offset grows through the programme, from roughly +90 near the front to past +130 by the Need Scheduler – it is not a flat shift, so each reference was re-checked individually against the 3.3 Program Structure Overview, not just incremented). Two items previously left open in Rev. 2 – the master Oil Controller enable (M65) and the flowmeter fitted/not-fitted detach – are now formally documented in the Program 3.3 PLC Guide (as M65 and the M800 toggle respectively) and have been rewritten below to match the confirmed ladder behaviour rather than flagged as open. Four genuinely new checklist items have been added for functionality introduced in 3.3: the fault trigger-value snapshot (D750–D768, “max value recorded”), the T250/T251 dispense-timer renumbering, the M800 flowmeter toggle's exact forced value, and the Manual Mode Scheduler/Manual Pump Mode consolidation – see the highlighted items throughout, and Section 5 for all five collected in one place. One item from Rev. 2 remains open: M51 (Run-or-Pause) also withholding a paused tank from priority scheduling is still not written into the Program 3.3 PLC Guide's own walkthrough or register tables, despite the renumbering. See Open Items.

1 HMI (TK8100H)

1.1 Home Screen

Verification Item	Expected Result	Tolerance	P	F	U
T5 trend vs. thermal potential	Home screen trend graph plots Temp T5 (D64) against the calculated Dynamic Upper Thermal Boundary T_f (D296), refreshing every T0 pulse (~ 1 s), sourced from the four-point HMI trend mirror (D600–D603) built in Charting (PLC Step 3277).	$\pm 5\%$	☐	☐	☐
Full trend set present	The same trend mirror also carries T_{target} (D130 $\rightarrow$ D602) and Level T5% (D76 $\rightarrow$ D603); the Home screen shows all four channels updating together, with no frozen or missing traces.	Exact	☐	☐	☐

Graph axis scaling was tested for in Rev. 1; confirmed with the design authority that exact scaling behaviour is not something this checklist needs to verify, so the item has been removed.

1.2 Water Circulation

Verification Item	Expected Result	Tolerance	P	F	U
Manual valve operation	With M12 (manual mode) active, each branch valve output (Y011–Y014, Scheduler Gates) responds directly to its HMI manual pushbutton.	Exact	☐	☐	☐
Mode adjusts on gating	Circulation valve pattern (Y011–Y014) follows the active Scheduler Gate (M100–M109) selected by the LSM/HSM/FSM cascade (PLC Step 2538).	Exact	☐	☐	☐
Mode displayed	HMI mode field shows “FSM”, “HSM”, or “LSM” as written to D480 by the Mode Indicator (PLC Step 1868).	Exact	☐	☐	☐
No button response in Auto	With M11 (auto) set and M12 (manual) clear, the Water Circulation screen’s manual valve pushbuttons produce no output change.	Exact	☐	☐	☐

1.3 PO Tank

Verification Item	Expected Result	Tolerance	P	F	U
Designation displayed	Tank-select flags M14/M15/M16 (PO Designation, PLC Step 1349) are reflected as the designated tank(s) on the PO Tank screen.	Exact	☐	☐	☐
Enabling requires password	Starting a PO dispense sequence (M50 Initiate) is gated by an HMI-side password prompt. The PLC's M50/M51/M52 button trio (PO Dispensing, PLC Step 1587) carries no password logic itself – the gate is enforced entirely on the HMI.	Exact	☐	☐	☐
Tank score displayed	PO Tank screen shows the live weighted score (D330/D332/D334, Tank Scoring, PLC Step 1605) for tanks A/B/C.	±5%	☐	☐	☐
Start button resets correctly	M50 (Initiate) clears only when the PO-melt-demand watchdog – renumbered from T5 to T250 in this export – times out on completion ($D340 = 0$, where $D340 = D302 \times 10$ is T250's own preset) or when M52 (Reset) is pressed – it does not clear mid-cycle.	Exact	☐	☐	☐
Pause stops timer and priority	M51 (Run-or-Pause), part of the Initiate/Run-or-Pause/Reset trio (PLC Step 1587), both halts the PO watchdog timer T250 (renumbered from T5) and withholds the paused tank from priority evaluation in the Need/Mode Scheduler – a deliberate addition confirmed by the design authority, still not reflected in the PLC Guide's own walkthrough or register tables in the 3.3 edition.	Exact	☐	☐	☐

1.4 IO Screen

Verification Item	Expected Result	Tolerance	P	F	U
Input/output indicators	Each labelled X (input) and Y (output) indicator on the IO screen energises exactly when its physical bit is true. Inputs mirror through M115–M146 (HMI Display for Inputs, PLC Step 3143).	Exact	☐	☐	☐

1.5 Sensor Screen

Verification Item	Expected Result	Tolerance	P	F	U
Sensor name mapping	Each displayed sensor tag matches the confirmed channel order (index 0 = Temp T5 ... index 9 = Level TC; RTD/Ultrasonic scaling, PLC Steps 223/266) – e.g. the tag reading “T5” shows the T5 channel value, not TA's.	Exact	☐	☐	☐

1.6 Sensor Configuration

Verification Item	Expected Result	Tolerance	P	F	U
Sensor value matches name	As per the Sensor Screen check, the value shown against each channel's configuration entry belongs to that channel and no other.	Exact	☐	☐	☐
Fault limit registered	Each channel's adjustable fault threshold (D240–D278, Error Threshold, PLC Step 614) is visible and editable per channel on the Sensor Configuration screen.	Exact	☐	☐	☐
Max value recorded display	<i>New in 3.3.</i> For each of the ten monitored channels, when the rate-of-change value drops back below its second (lower) fault threshold, the value at that moment is captured into the channel's trigger-value snapshot register (D750 channel 0 through D768 channel 9, Error Threshold, PLC Step 614) and shown on the Sensor Configuration screen as “max value recorded”. Force a fault, let it clear, and confirm the displayed figure matches the rate-of-change value at the instant of clearing, not the live value afterwards.	±5%	☐	☐	☐
Detach flowmeter also works	The flowmeter fitted/not-fitted toggle (M800 , an HMI switch) selects whether a physical flowmeter is installed at this site. With M800 set, the processed flow register (D42) is forced to a fixed nominal value of 1 – not a site-measured placeholder as in the 3.2 edition – letting the Oil Distribution timing maths (PLC Step 2843) run without a live flow signal; with M800 clear, D42 takes its normal scaled value from Flowmeter to RPM (PLC Step 309). Confirm the forced value holds at exactly 1 while set, and that oil-distribution calculations downstream continue to run without fault in either state.	Exact	☐	☐	☐

1.7 Heating Screen

Verification Item	Expected Result	Tolerance	P	F	U
Target/current water graph	T_{target} (D130) and Temp T5 (D64) trend traces (mirrored to D602/D601 in Charting, PLC Step 3277) update every T0 pulse and settle without dropouts as the quadratic ramp (Water Temperature Target Logic, PLC Step 915) converges.	±5%	☐	☐	☐

1.8 Oil Distribution

Verification Item	Expected Result	Tolerance	P	F	U
Master switch limits all flow	M65 is the master Oil Controller enable, AND-gated alongside M11/M200/X014 into the dispense rung (Oil Distribution Timing, PLC Step 2959). The 3.3 PLC Guide now confirms M65 is sourced specifically from the TK8100H supervisory HMI , independent of the three per-station TK8050H units. De-asserting M65 blocks Y005 for every station regardless of lock/timer state; asserting it restores normal dispensing. Confirm separately that toggling controls on a per-station TK8050H screen has no effect on M65.	Exact	☐	☐	☐
Flowrate displayed	Processed flow reading (D42, Flowmeter to RPM, PLC Step 309) is shown live on the Oil Distribution screen; with the M800 flowmeter-not-fitted toggle set, this should read a fixed 1 rather than a live value (cross-ref Sensor Configuration, Detach flowmeter also works).	$\pm 5\%$ (fitted) / Exact (not fitted)	☐	☐	☐
Calibration toggling available	The dispense-rate parameters (D500/D502, Oil Distribution Calculation, PLC Step 2843) are HMI-exposed and editable, gated by password. Confirm the password prompt appears before the fields become editable, and that a changed value takes effect on the next dispense cycle.	Exact (access) / $\pm 5\%$ (applied value)	☐	☐	☐
Oil usage chart and reset	Daily per-station totals (D604–D606) accumulate on each completed dispense and zero cleanly on the 24-hour rollover (D610 vs. K1440, Oil Chart, PLC Step 3207).	$\pm 5\%$ value / Exact reset timing	☐	☐	☐

2 Control Panel Buttons

Verification Item	Expected Result	Tolerance	P	F	U
E-Stop hardware trip	With M60 in its normal (enabled) state, pressing the physical E-Stop (X000) immediately asserts M10 and the E-Stop output Y000 (E-Stops, PLC Step 43).	Exact	☐	☐	☐
E-Stop bypass access control	M60 can only be cleared – disabling the E-Stop path for maintenance – after the correct password is entered on the HMI. The PLC-side M60 coil carries no password logic itself; the gate is enforced entirely on the HMI.	Exact	☐	☐	☐
Auto/Manual selector	Auto position sets M11 and lights Y001; Manual position sets M12 (Automation, PLC Step 56). The two are mutually exclusive.	Exact	☐	☐	☐
Circulation pump	“Circulation pump” denotes no separate coil of its own – it is the circulation valve pattern (Y011–Y014) that auto-adjusts with whichever Scheduler Gate (M100–M109) the active LSM/HSM/FSM mode has opened (PLC Step 2538). Same test as Water Circulation, Mode adjusts on gating, confirmed from the control panel side.	Exact	☐	☐	☐
Oil pump	Confirmed: “Oil pump” is Y005, the oil dispense valve output for the active station, AND-gated with M65 (Oil Distribution Timing, PLC Step 2959).	Exact	☐	☐	☐
Jacket pump	Y006 runs unconditionally whenever M11 (auto) is set (Jacket Pump, PLC Step 3015); manual override via M12 AND X004.	Exact	☐	☐	☐
AODD pump	Y007 engages when M11 is set and any tank temperature final (D66/D68/D70) exceeds K30 (AODD Pump, PLC Step 3020); manual override via M12 AND X005.	Exact logic / $\pm 5\%$ threshold	☐	☐	☐
Heaters (stages 1–3)	Y010–Y012 follow the hysteretic stage counter D350 (Heater Latching, PLC Step 1038) in auto mode, or drive directly from pushbuttons X006/X007/X010 in manual mode (Manual Heater Control, PLC Step 1210).	Exact	☐	☐	☐
Manual mode-select buttons	<i>New in 3.3.</i> The 3.2 edition split the manual LSM/HSM/FSM mode-select override and the manual pump-mode pushbuttons into two separate named sections. In 3.3 both are consolidated under one heading, Manual Pump Mode (PLC Step 3053): confirm the mode-select outputs (Y002–Y004, via M304–M306) and the pump-mode pushbuttons (M300–M303) both still operate correctly from this single section, with no change to which physical buttons drive them.	Exact	☐	☐	☐

3 Math

Every calculated quantity below is checked the same way: I independently hand-calculate the expected value from the same live inputs the PLC is reading, then compare it to the PLC's own register value. A result within $\pm 5\%$ of the hand-calculated value is a Pass. Placeholder constants (Section 3) are tested against whatever value is actually loaded at the time of the test, not against a final site-calibrated figure.

Verification Item	Expected Result	Tolerance	P	F	U
RTD/Ultrasonic/Flow scaling	Processed values (D22–D42) match DEMUL E2.5 / DESUB E1 of the corresponding raw channel (PLC Steps 223–336).	$\pm 5\%$	☐	☐	☐
5-sample sensor average	Final average (D64–D84) matches a 5-sample mean of the processed channel, resetting every 5 s (Sensor Averaging, PLC Steps 336–366).	$\pm 5\%$	☐	☐	☐
Level % conversion	Level % output (D122–D128) matches $\left(\frac{100 - \text{level}}{100}\right) \times 100$ against the live level final (Level % Converter, PLC Step 405).	$\pm 5\%$	☐	☐	☐
Rate-of-change ratio	Ratio (D220–D238) matches the live final divided by the 11 s-old snapshot (D200–D218), per channel (PLC Step 584).	$\pm 5\%$	☐	☐	☐
T_{target} quadratic ramp	D130 matches $70 - \frac{3(D122-50)^2}{160}$ for $50 \leq D122 \leq 90$, flat at 70.0/40.0 outside that band (Water Temperature Target Logic, PLC Step 915).	$\pm 5\%$	☐	☐	☐
Dynamic Upper Thermal Boundary	T_f (D296) matches the mass-weighted mixing formula using Temp SOL (D74) and the T5 mass proxy (D280) against Temp T5 (D64), constant E239.53 (PLC Step 1273).	$\pm 5\%$	☐	☐	☐
Height-to-mass proxy	Mass proxies (D280–D286) match level final $\times K1$ (currently unity gain – placeholder, PLC Step 1220).	$\pm 5\%$ of the loaded gain, not final calibration	☐	☐	☐
Tank score weighting	Tank score (D330/D332/D334) matches $0.6 \times \text{mass proxy} + 0.4 \times \text{temperature final}$ for that tank (Tank Scoring, PLC Step 1605).	$\pm 5\%$	☐	☐	☐
PO melt/liquid demand	D302 matches the two-branch demand function against the K30 melting threshold (PO Timer, PLC Step 1410).	$\pm 5\%$	☐	☐	☐
Oil dispense duration	D316 matches the MODBUS-read value, processed flow, rate parameter D500, and exponential estimate combined per Oil Distribution Calculation (PLC Step 2843). Note the value actually loaded into the dispense timer is $D316 \times 10$, feeding T251 (renumbered from T100) – verify D316 itself against this tolerance, and separately confirm the timer's live preset reads exactly $10 \times D316$.	$\pm 5\%$ (D316) / Exact ($\times 10$ relationship)	☐	☐	☐

4 Alarms and Safety

All ten rate-of-change fault bits (M0–M9) are checked against the Error Threshold logic (PLC Step 614), which runs on the T4 (11 s) pulse. Channel order follows the confirmed sensor index (Section 4, PLC Step 223/266): index 0 = Temp T5 through index 9 = Level TC. The flow channel (index 10) is not alarmed here.

Verification Item	Expected Result	Tolerance	P	F	U
M0 – Temp T5 fault	M0 asserts when Temp T5's rate-of-change ratio (D220) crosses its threshold (D240/D242); HMI sounds an alarm and shows a banner for this channel.	Exact assert / $\pm 5\%$ threshold	☐	☐	☐
M1 – Temp TA fault	As above, for D222 vs. D244/D246.	Exact / $\pm 5\%$	☐	☐	☐
M2 – Temp TB fault	As above, for D224 vs. D248/D250.	Exact / $\pm 5\%$	☐	☐	☐
M3 – Temp TC fault	As above, for D226 vs. D252/D254.	Exact / $\pm 5\%$	☐	☐	☐
M4 – Temp HEA fault	As above, for D228 vs. D256/D258.	Exact / $\pm 5\%$	☐	☐	☐
M5 – Temp SOL fault	As above, for D230 vs. D260/D262.	Exact / $\pm 5\%$	☐	☐	☐
M6 – Level T5 fault	As above, for D232 vs. D264/D266.	Exact / $\pm 5\%$	☐	☐	☐
M7 – Level TA fault	As above, for D234 vs. D268/D270.	Exact / $\pm 5\%$	☐	☐	☐
M8 – Level TB fault	As above, for D236 vs. D272/D274.	Exact / $\pm 5\%$	☐	☐	☐
M9 – Level TC fault	As above, for D238 vs. D276/D278.	Exact / $\pm 5\%$	☐	☐	☐
Alarm acknowledge / silence	The HMI alarm banner and beep persist until either the rate-of-change ratio returns within its normal bounds (the corresponding Mx bit clears on its own, Error Threshold, PLC Step 614) or M60 is pressed to force a shutdown – there is no separate silence/mute action; the alarm only stops on fault clearance or E-Stop.	Exact	☐	☐	☐
E-Stop on persistent fault	A persistent fault that will not clear can be escalated to a full shutdown via the M60 HMI override, itself gated by password (see Control Panel Buttons, E-Stop bypass access control).	Exact	☐	☐	☐

5 New Features Verification – Program 3.3

This section collects, in one place, the five items genuinely new in Program 3.3 (as opposed to renumbered). The corresponding test is also present in its normal HMI/Control Panel/Math section above – use either location to tick it off, but it only needs to be signed once. Use this section as the fast, standalone check when re-verifying after a 3.2 → 3.3 upgrade on an already-commissioned line.

New Feature	Expected Result	Tolerance	P	F	U
Fault trigger-value snapshot (D750–D768)	Per-channel “max value recorded” captures the rate-of-change value at the instant the channel’s lower fault threshold is crossed on the way down (Error Threshold, PLC Step 614). See Sensor Configuration, Max value recorded display.	±5%	☐	☐	☐
T250/T251 dispense-timer renumbering	PO-melt watchdog (was T5) and oil-dispense timer (was T100) now run as Coolmay 100 ms-base devices T250 and T251, loaded from D340 (=D302×10) and D316×10-in-place respectively. Confirm both device numbers directly on the PLC programming software and that behaviour (start/clear/reset conditions) is otherwise unchanged. See PO Tank, Start button resets correctly / Pause stops timer and priority; Math, Oil dispense duration.	Exact (device identity)	☐	☐	☐
M800 flowmeter-fitted toggle	HMI switch forcing processed flow (D42) to a fixed nominal value of 1 when set (flowmeter not fitted), vs. its live scaled value when clear. See Sensor Configuration, Detach flowmeter also works; Oil Distribution, Flowrate displayed.	Exact	☐	☐	☐
M65 master Oil Controller enable	Confirmed sourced from the TK8100H supervisory HMI specifically, independent of the three per-station TK8050H units; gates Y005 system-wide alongside M11 and the active station lock. See Oil Distribution, Master switch limits all flow; Control Panel Buttons, Oil pump.	Exact	☐	☐	☐
Manual Pump Mode consolidation	The separate 3.2 “Manual Mode Scheduler” and “Manual Pump Mode” sections are now one section (PLC Step 3053) driving both the mode-select outputs and the pump push-buttons. See Control Panel Buttons, Manual mode-select buttons.	Exact	☐	☐	☐

Open Items

One item remains outstanding as of Rev. 3. Of the three items carried forward from Rev. 1 into Rev. 2 as ladder behaviour confirmed by the design authority but not yet written into the PLC Guide, two – the master Oil Controller enable (M65) and the flowmeter fitted/not-fitted detach (now M800) – are formally documented in the Program 3.3 PLC Guide and have been closed out above.